



Developing the integral framework

- We introduce particle filters in the context of estimating the conditional mean of a nonlinear system's state:

$$\mathbb{E}[x_k | \mathbb{Z}_k] = \int_{R_{x_k}} x_k f(x_k | \mathbb{Z}_k) dx_k,$$

where R_{x_k} is the set comprising the range of possible x_k .

- Previously, the Gaussian assumption allowed us to derive linear KF steps in closed form, which EKF and SPKF approximated in different ways for nonlinear systems.
- Now, however, we no longer assume that all probability densities are Gaussian.
- So, we must develop a new approach: one that allows us to approximate this integral directly within a practical computational framework.
- Ultimately, we will estimate $f(x_k | \mathbb{Z}_k)$, which is more informative than $\mathbb{E}[x_k | \mathbb{Z}_k]$.



Invoking Bayes' rule

- To find $f(x_k | \mathbb{Z}_k)$, we first break it into simpler pieces.
- We start by invoking Bayes' rule to rewrite $f(x_k | \mathbb{Z}_k)$:

$$f(x_k | \mathbb{Z}_k) = \frac{f(\mathbb{Z}_k | x_k) f(x_k)}{f(\mathbb{Z}_k)}.$$

- We then write $\mathbb{Z}_k$ in terms of its constituent parts $\mathbb{Z}_{k-1}$ and z_k :

$$\frac{f(\mathbb{Z}_k | x_k) f(x_k)}{f(\mathbb{Z}_k)} = \frac{f(z_k, \mathbb{Z}_{k-1} | x_k) f(x_k)}{f(z_k, \mathbb{Z}_{k-1})}.$$

- Next, we apply two versions of the joint-probability rules $f(a, b) = f(a | b) f(b)$ and $f(a, b | c) = f(a | b, c) f(b | c)$ to the numerator and denominator terms:

$$\frac{f(z_k, \mathbb{Z}_{k-1} | x_k) f(x_k)}{f(z_k, \mathbb{Z}_{k-1})} = \frac{f(z_k | \mathbb{Z}_{k-1}, x_k) f(\mathbb{Z}_{k-1} | x_k) f(x_k)}{f(z_k | \mathbb{Z}_{k-1}) f(\mathbb{Z}_{k-1})}.$$



Invoking Bayes' rule again, simplifying

- We apply Bayes' rule once again in the terms within the []:

$$\begin{aligned} & \frac{f(z_k | \mathbb{Z}_{k-1}, x_k) [f(\mathbb{Z}_{k-1} | x_k)] f(x_k)}{f(z_k | \mathbb{Z}_{k-1}) f(\mathbb{Z}_{k-1})} \\ &= \frac{f(z_k | \mathbb{Z}_{k-1}, x_k) [f(x_k | \mathbb{Z}_{k-1}) f(\mathbb{Z}_{k-1})] f(x_k)}{f(z_k | \mathbb{Z}_{k-1}) f(\mathbb{Z}_{k-1}) [f(x_k)]}. \end{aligned}$$

- Next, we cancel some common terms from the numerator and denominator:

$$\frac{f(z_k | \mathbb{Z}_{k-1}, x_k) f(x_k | \mathbb{Z}_{k-1}) f(\mathbb{Z}_{k-1}) f(x_k)}{f(z_k | \mathbb{Z}_{k-1}) f(\mathbb{Z}_{k-1}) f(x_k)} = \frac{f(z_k | \mathbb{Z}_{k-1}, x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- Finally, we recognize that z_k is conditionally independent of $\mathbb{Z}_{k-1}$ given x_k :

$$\frac{f(z_k | \mathbb{Z}_{k-1}, x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})} = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$



Conditionally independent?

- The prior slide said that “ z_k is conditionally independent of $\mathbb{Z}_{k-1}$ when given x_{k-1} .” What does this mean?
- Recall that our nonlinear state-space system model has the form:

$$\begin{aligned}x_k &= f(x_{k-1}, u_{k-1}, w_{k-1}) \\ z_k &= h(x_k, u_k, v_k).\end{aligned}$$

- The present measurement z_k is a direct function of x_k (and u_k and v_k). If I am given x_k , $\mathbb{Z}_{k-1}$ gives no additional information when evaluating $f(z_k | x_k, \mathbb{Z}_{k-1})$.
- Similarly (useful on next slide) the state equation shows that if I am given x_{k-1} , then $\mathbb{Z}_{k-1}$ gives no additional information when computing $f(x_k | x_{k-1}, \mathbb{Z}_{k-1})$.
 - Therefore, “ x_k is conditionally independent of $\mathbb{Z}_{k-1}$ when given x_{k-1} .”



How to compute $f(x_k | \mathbb{Z}_{k-1})$?

- Overall, to this point, we have shown that:

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- To proceed further using this approach, we require ways to compute the relevant conditional probability densities.
- For example, we can apply a version of the law of total probability:

$$f(a | b) = \int_{R_c} f(a | b, c) f(c | b) dc$$

to find a different form for $f(x_k | \mathbb{Z}_{k-1})$, where “ c ” is x_{k-1} :

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1},$$

remembering that x_k is conditionally independent of $\mathbb{Z}_{k-1}$ when given x_{k-1} .



Key point #1

- **KEY POINT #1:** We can compute $f(x_k | \mathbb{Z}_k)$ recursively with two steps per iteration:
 - Step 1 computes probability densities for predicting x_k given all past observations:

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1}.$$

- Step 2 updates the prediction via:

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- Therefore, the general sequential-inference solution breaks naturally into a prediction/update scenario.
- We “just” need to find $f(x_k | x_{k-1})$ and $f(z_k | x_k)$. Note: $f(z_k | \mathbb{Z}_{k-1})$ is a normalizing constant that does not need to be computed explicitly.



How to compute $f(x_k | x_{k-1})$ and $f(z_k | x_k)$?

- We compute the two remaining conditional densities using the law of total probability. We can write:

$$f(x_k | x_{k-1}) = \int_{R_{w_k}} \delta(x_k - f(x_{k-1}, u_{k-1}, w_{k-1})) f(w_{k-1}) dw_{k-1} = \sum f(w_{k-1}),$$

where the summation is over all $\{w_{k-1} : x_k = f(x_{k-1}, u_{k-1}, w_{k-1})\}$

- Similarly,

$$f(z_k | x_k) = \int_{R_{v_k}} \delta(z_k - h(x_k, u_k, v_k)) f(v_k) dv_k = \sum f(v_k),$$

where the summation is over all $\{v_k : z_k = h(x_k, u_k, v_k)\}$.

- Idea: Sum probabilities of all values of process noise w_{k-1} that could move the state from x_{k-1} to x_k and sum probabilities of all values of sensor noise v_k that could produce the output z_k when the state is x_k .



Key point #2

- **KEY POINT #2:** Closed-form solutions to solve these multi-dimensional integrals are intractable for most real systems.
 - A simplified solution may be obtained if we are willing to make the *assumption* that all probability densities are *Gaussian*.
 - This was the basis for the original Kalman filter, the extended Kalman filter, and the sigma-point (unscented) Kalman filter.
 - They can actually be derived from this starting point with the Gaussian assumption in a little different way from what we looked at earlier.
 - For applications that justify the computational expense, the integrals may be approximated computationally using Monte-Carlo methods.
 - This is what particle filters do, which is the focus of this course.



Summary

- We desire to find a way to evaluate metrics like:

$$\mathbb{E}[x_k | \mathbb{Z}_k] = \int_{R_{x_k}} x_k f(x_k | \mathbb{Z}_k) dx_k.$$

- Ultimately, we will estimate $f(x_k | \mathbb{Z}_k)$, which is more informative than $\mathbb{E}[x_k | \mathbb{Z}_k]$.
- We recursively estimate $f(x_k | \mathbb{Z}_k)$ using two steps.
 - Step 1 computes probability densities for predicting x_k given all past observations:

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1}.$$

- Step 2 updates the prediction via:

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- Particle filters use computational means to estimate the integrals in these two steps.